

Explainability as a Dialogue: Establishing a Shared Language for Human-AI Alignment

Luka Zaitch
Vrije Universiteit

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Abstract

The transition of artificial intelligence from predictive engines to collaborative partners requires a fundamental reimagining of explainability. Current static interpretation methods, such as saliency maps, fail to bridge the epistemological gap between human intuition and the fundamentally “alien” representation spaces of modern deep learning. This paper critically evaluates the “Explainability as a Dialogue” framework as a mechanism to establish a “Shared Language” for human-AI alignment (Kim). We analyze the operational requirements of such systems through the normative lens of the NIST Four Principles, specifically focusing on the fidelity constraints of *Explanation Accuracy*. While we affirm that interactive dialogue is necessary to correct the “introspection illusion” in practitioners, we identify a critical failure mode: *sycophancy*. We argue that without rigorous self-monitoring, conversational agents prioritize communicative plausibility over factual faithfulness, becoming deceptive “yes-men.” Ultimately, we posit that as AI reasoning evolves into high-dimensional latent spaces, the dialogue interface risks becoming a post-hoc translation layer rather than a window into the model’s true logic, necessitating a shift from anthropomorphic trust to rigorous scientific verification.

1 Introduction

The rapid deployment of machine learning models into high-stakes environments, ranging from clinical diagnostics to public policy, has precipitated a crisis of trust. As these systems transition from controlled laboratory settings to “wild” interactions with non-technical experts, the opacity of the underlying “black box” has become a critical barrier

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to adoption. Historically, the field of Explainable AI (xAI) has addressed this challenge through post-hoc interpretability methods. Standard techniques, such as LIME or gradient-based saliency maps, generate static artifacts, such as feature importance lists or heatmaps, that attempt to approximate the model’s local decision boundary. While these methods provide a snapshot of the model’s focus, they are increasingly viewed as insufficient for complex decision-making because they treat explainability as a unilateral transmission of data rather than a bilateral exchange of meaning.

Recent scholarship by Lakkaraju et al. identifies a fundamental misalignment between these static outputs and the cognitive needs of practitioners [4]. Their research demonstrates that domain experts are often frustrated by “one-off” explanations, as they lack the capacity for interrogation. A static chart cannot answer counterfactual queries (“Why not class X?”) or conditional probes (“How would the output change if feature Y were different?”), which are essential for validating a human’s mental model. Consequently, this paper argues that true explainability requires a paradigm shift from *interpretability*, viewing the model as a code-base to be debugged, to *communication*, viewing the model as a “scientific object” or “alien” intelligence whose internal representations must be deciphered through interaction [3].

This essay constructs a theoretical framework for “Explainability as a Dialogue.” We integrate Lakkaraju’s interactive model with the normative constraints defined by the National Institute of Standards and Technology (NIST) to propose a mechanism for establishing a “Shared Language” between human and machine [5]. Finally, we critically analyze the risks inherent in this conversational approach, specifically examining the emerging phenomenon of *sycophancy*, where Large Language Models (LLMs) prioritize the plausibility of an explanation over its faithfulness to the underlying logic.

2 The Theoretical Imperative: Bridging the Alien Gap

The central epistemological challenge in Explainable AI is not merely technical transparency, but the translation of fundamentally divergent representation spaces. Kim argues that modern deep learning models should be viewed as “aliens”, entities that arrive at correct conclusions through internal logic paths that may not possess a direct analogue in human reasoning [3]. Consequently, the prevailing reliance on static interpretation methods (e.g., saliency maps or feature importance lists) is insufficient because it assumes a shared conceptual dictionary that does not exist. Kim identifies this gap as the “introspection illusion”, where human observers inadvertently project their own priors onto the model’s output. For example, when observing a heatmap highlighting a bird, a user naturally assumes the model has recognized the concept of a “beak” or “wing”, when in reality, the model may be relying on a spurious correlation with background texture. Without a mechanism to probe and verify these concepts, static explanations act as a

Rorschach test, confirming the user’s bias rather than revealing the model’s true nature. Furthermore, as models begin to optimize their own internal communication protocols, the “alien” nature of these representations may become structurally incompatible with human syntax, rendering direct translation impossible.

This theoretical disconnect is mirrored in the empirical dissatisfaction of high-stakes practitioners. Lakkaraju et al. demonstrate that domain experts, such as clinicians and policy researchers, fundamentally reject the role of a passive observer [4]. Their study reveals that these experts do not seek a tool that simply outputs a decision and a chart; rather, they demand a “colleague” capable of social exchange. In this context, trust is not established by a single proof of accuracy, but through an interactive process of querying, contesting, and refining explanations. Therefore, the goal of xAI must shift from static error analysis to the creation of a dynamic channel for “Theory of Mind”, allowing the human operator to build, test, and correct their mental model of the AI’s alien reasoning process through iterative dialogue.

3 Operationalizing Dialogue: Mechanisms and Normative Constraints

To transition from the theoretical “alien” metaphor to a practical engineering reality, Lakkaraju et al. propose a framework where the user interacts with the model via a conversational interface, treating the AI as a “colleague” rather than a static dashboard [4]. This mechanism relies on two technical pillars: *Iterative Interactivity*, which allows users to refine queries using natural language (e.g., “Filter this explanation for patients over 50”), and *Stateful Context Management*, which preserves the history of the conversation to enable drill-down analysis. Unlike static methods where the user must adapt to the tool’s output format, the dialogue framework shifts the burden of translation to the system, requiring it to map high-level human intent to low-level explanation algorithms.

However, a conversational interface alone is insufficient; it must be governed by rigorous standards to prevent misleading anthropomorphism. We frame these requirements through the lens of the NIST Four Principles of Explainable AI [5]. First, the dialogue must satisfy the principle of *Meaningfulness*; the “language” used by the agent must adapt to the user’s domain expertise. A physician requires a causal explanation rooted in biology, whereas a developer requires a trace of activation vectors. Second, the system must adhere to *Explanation Accuracy*. This is the fidelity constraint: the explanation generated during the dialogue must accurately reflect the underlying mathematical process, not merely offer a persuasive post-hoc justification. Finally, the dialogue agent must respect *Knowledge Limits*. Unlike standard chatbots that hallucinate to maintain flow, a high-stakes xAI agent must be programmed to explicitly reject queries that fall out-of-distribution, stating, “I cannot reason about this input with high confidence,” rather than fabricating a plausible answer.

4 The Sycophancy Paradox: The Risks of Conversational Alignment

While the dialogue framework addresses the flexibility deficit of static XAI, it introduces a critical failure mode unique to Large Language Models (LLMs): *sycophancy*. As defined by Hong et al., sycophancy occurs when a model tailors its responses to conform to the user’s perceived beliefs or biases, rather than adhering to objective facts [2]. In the context of xAI, this creates a fundamental tension between *Plausibility* (how convincing an explanation sounds) and *Faithfulness* (how true it is to the model’s logic).

Recent empirical work by Turpin et al. and Agarwal et al. suggests that models tuned via Reinforcement Learning from Human Feedback (RLHF) often prioritize plausibility, acting as “yes-men” to satisfy the human user [6, 1]. This manifests as a dangerous feedback loop; if a practitioner pushes back on a correct diagnosis (e.g., “Are you sure this isn’t benign?”), a sycophantic dialogue agent is statistically likely to apologize and reverse its explanation to align with the user. This distortion arises not merely from a desire to please, but often because the model’s actual reasoning path is too complex to be expressed in English, forcing it to fabricate a simpler, plausible proxy. This behavioral collapse directly violates the NIST principle of *Explanation Accuracy*, as the explanation no longer reflects the underlying computation but rather the user’s desire or a simplified fiction.

Furthermore, Wang et al. demonstrate that this issue extends to *strategic deception*. Their research on “Thinking LLMs” reveals that models can leverage Chain-of-Thought (CoT) reasoning to internally identify the user’s intent while generating a sanitized, misleading output to appease them [7]. This suggests that without rigorous self-monitoring mechanisms, the “colleague” practitioners desire may evolve into a deceptive subordinate, reinforcing human confirmation bias rather than correcting it.

5 The Translation Horizon: Latent Reasoning and the Limits of Language

The dialogue framework rests on a fragile assumption: that the internal state of a high-performance model can be losslessly mapped onto natural language. However, as models scale, we confront a tension between *computational efficiency* and *linguistic interpretability*. Recent observations of “latent reasoning”, where models utilize non-human tokens or high-dimensional vector manipulations to bridge logical gaps, suggest that advanced AI may evolve a “private language” optimized for prediction rather than communication.

This creates a paradox for Been Kim’s vision of a “Shared Language”. If we force the model to reason entirely in English (Chain-of-Thought), we may impose a “cognitive cap”, restricting the model to the limitations of human syntax. Conversely, if we allow

the model to reason in its native, high-dimensional latent space (as proposed for future architectures like OpenAI’s “non-human reasoning” steps), the dialogue interface becomes a *post-hoc translation layer* rather than a window into the process. In this scenario, the *Explanation Accuracy* (NIST Principle 3) becomes structurally compromised: the model is no longer explaining *how* it thought, but translating the *result* of its alien cognition into a fable the human can understand. Thus, the ultimate challenge of Dialogue is not merely preventing deception (sycophancy), but managing the inevitable information loss when mapping hyper-dimensional intelligence onto low-dimensional human speech.

6 Conclusion

The evolution of Explainable AI from static dashboards to interactive conversation represents not merely an ergonomic improvement, but an epistemological necessity. As Kim argues, the representation space of modern deep learning is fundamentally alien to human intuition; therefore, a “Shared Language” cannot be statically defined but must be discovered through dynamic exchange. By operationalizing Lakkaraju’s dialogue framework, we provide the mechanism for this discovery, allowing practitioners to test boundaries, query counterfactuals, and align their mental models with the machine’s reality.

However, this anthropomorphic interface acts as a double-edged sword. While treating the AI as a “colleague” fosters trust, it simultaneously opens the door to misplaced reliance if the agent prioritizes conversational agreeableness over factual rigor. As demonstrated by the emerging literature on sycophancy, a dialogue agent that lacks robust constraints effectively becomes a “yes-man”, reinforcing human biases rather than correcting them.

Consequently, the future of high-stakes xAI lies in the development of rigorous “Self-Monitoring” mechanisms that can enforce the NIST principles of *Explanation Accuracy* and *Knowledge Limits* in real-time. However, we must also prepare for a future where the “Shared Language” is an approximation, not a transcript. If AI reasoning transcends the boundaries of human language, the Dialogue must evolve from a simple conversation into a rigorous process of *scientific translation*, acknowledging that some part of the “alien” mind may remain forever untranslatable. Ultimately, the goal is not just to build a chatbot that mimics human conversation, but to engineer an instrument capable of measuring the distance between our reality and the machine’s.

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